Enterprise Parking Lot LLD Specification

Version: 4.0 (Production Ready)
Architecture Variant: Highly Concurrent, Fault-Tolerant Micro-Engine

1. System Overview

This document specifies the structural LLD blueprint for an enterprise-grade multi-threaded Parking Management System. The architecture resolves critical edge cases around high-concurrency race conditions, deadlock risks in tight drive lanes, dynamic yield-management pricing structures, and real-time prioritizations for premier clientele.

2. Advanced Architectural Additions

- **Multi-Currency Crypto Gateway Strategy:** Decouples volatile settlement tasks away from core threads via a decoupled Strategy Pattern layout. Processes high-throughput, cross-currency payments with minimum locking latency.
- **Valet Routing Optimization Matrix:** Uses a multi-tiered graph structure to dynamically sort floor access layout sequences, minimizing vehicle driving transit time based on live physical distance weights.
- **Deadlock Mitigation Lane Engine:** Wraps physical structure entry corridors in fair-ordered ReentrantLock barriers with non-blocking tryLock limits to prevent cascading traffic gridlocks.
- **Automated EV Charging Scheduler:** Monitors live grid limitations across active parking spots via an internal load balance calculation layout to optimize electrical distribution constraints.

3. Thread-Safety Matrix & Component Mapping

Component / Class	Concurrency Control Primitive	Operational Function
ParkingSystem	ConcurrentHashMap, Locks	Orchestrates isolated entry lanes and lock-free departures.
ParkingLot	AtomicInteger	Tracks real-time load thresholds across the property without locks.
ParkingSpot	Intrinsic Locks (synchronized)	Guarantees single atomic reservation assignments.
ReservationSystem	PriorityBlockingQueue	Maintains high-priority premium queues across caller pools.
EvChargingScheduler	ReentrantLock	Sequences electrical draws preventing power supply spikes.
ParkingAnalyticsEngine	LongAdder	Aggregates metric throughput counter matrix with zero contention.

4. Automated EV Charging Scheduler Specification

The EV Charging Scheduler uses a thread-safe state distribution model to track active grid draws. When a vehicle plugs into an EV-equipped regular/compact spot, the engine evaluates available load capacity down to the exact kilowatt. If charging requests exceed safe distribution limits, the scheduler automatically queues low-priority tiers, preventing brownouts while maintaining peak utility for premium tiers.

Disclaimer: This document is an architecture specification artifact generated for low-level software implementation tracking purposes.